

| Question number | Answer                                                                                                                                                                                                    | Mark |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 6(a)(i)         | <p>The only correct answer is <b>D</b>.</p> <p><i>A is incorrect because R is a glycoprotein</i><br/> <i>B is incorrect because Q is an enzyme</i><br/> <i>C is incorrect because S is the capsid</i></p> | (1)  |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                 | Mark |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 6(a)(ii)        | <p>The only correct answer is <b>D</b>.</p> <p><i>A is incorrect because Q is an enzyme, R is a glycoprotein and S is the protein capsid</i><br/> <i>B is incorrect because Q is an enzyme, R is a glycoprotein and S is the protein capsid</i><br/> <i>C is incorrect because Q is an enzyme, R is a glycoprotein and S is the protein capsid</i></p> | (1)  |

| Question number | Answer                                                                                                                                                                                                                                                      | Mark |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 6(a)(iii)       | <p>The only correct answer is <b>A</b>.</p> <p><i>B is incorrect because TMV does not have an envelope</i><br/> <i>C is incorrect because TMV does not have an envelope</i><br/> <i>D is incorrect because Ebola virus has both an envelope and RNA</i></p> | (1)  |

| Question number | Answer                                                                                                                                                                                                                                                                                                                   | Mark |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 6(a)(iv)        | <p>The only correct answer is <b>D</b>.</p> <p><i>A is incorrect because 60mm is 60 000 000 nm, divide this by 120nm = 500 000</i><br/> <i>B is incorrect because 60mm is 60 000 000 nm, divide this by 120nm = 500 000</i><br/> <i>C is incorrect because 60mm is 60 000 000 nm, divide this by 120nm = 500 000</i></p> | (1)  |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                         | Additional guidance                                                                                                                                                                                                                                                                                                                                                                   | Mark |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 6(b)(i)         | <p>An explanation that includes two of the following points:</p> <ul style="list-style-type: none"> <li>because the y axis would have to be extended to {37 / 38 / 39 / 40} (1)</li> <li>otherwise this would make the rest of the data too squashed (1)</li> </ul> <p>OR</p> <ul style="list-style-type: none"> <li>so that <u>all</u> the data can be plotted on one graph (1)</li> <li>so that it can <u>all</u> be compared (1)</li> </ul> | <p><b>ACCEPT</b> would require a bigger scale<br/>larger graph</p> <p><b>ACCEPT</b> changes in deaths and new infections would not be so accurately presented</p> <p><b>ACCEPT</b> with data for new infections and deaths<br/><b>IGNORE</b> easier / harder other data</p> <p><b>ACCEPT</b> with data for new infections and deaths<br/><b>IGNORE</b> easier / harder other data</p> | (2)  |

| Question number | Answer                                                                                                                                                                                                                                                                           | Additional guidance                           | Mark |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|------|
| 6(b)(ii)        | <p>An answer that includes two of the following points:</p> <ul style="list-style-type: none"> <li>two values read from graph, subtracted and divided by 5 (1)</li> <li><math>2 \times 10^5</math> / <math>2.0 \times 10^5</math> / <math>2.00 \times 10^5</math> (1)</li> </ul> | <p><math>(3.2 - 3.1) \div 5 = 0.02</math></p> | (2)  |

| Question number | Answer                                                                                                                                                                                                                                              | Additional guidance                                                                                                                        | Mark |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|------|
| 6(b)(iii)       | <p>An explanation that includes two of the following points:</p> <ul style="list-style-type: none"> <li>more infected people are surviving (1)</li> <li>(more) people are surviving due to {better / new} {treatments / health care} (1)</li> </ul> | <p><b>ACCEPT</b> fewer infected people are dying<br/> <b>IGNORE</b> death rates</p> <p><b>IGNORE</b> vaccines, antibiotics, technology</p> | (2)  |

| Question number | Answer                                                                                                                      | Additional guidance                                                                                          | Mark |
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| 6(b)(iv)        | <ul style="list-style-type: none"> <li>extrapolation (of data / graph / line) / extending the line (to 2025) (1)</li> </ul> | <p><b>ACCEPT</b> <u>extend</u> a line of best fit<br/> <b>IGNORE</b> draw a line of best fit unqualified</p> | (1)  |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Additional guidance                                                                   | Mark |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|------|
| 6(b)(v)         | <p>An explanation that includes two of the following points:</p> <ul style="list-style-type: none"> <li>because vaccines result in {(artificial active) immunity / an immune response} (1)</li> <li>due to the {formation / presence} of memory cells (1)</li> <li>therefore (as a result of the vaccine) there would be fewer people with HIV (1)</li> <li>and therefore non-immune people less likely to become infected by someone with HIV (1)</li> </ul> | <p><b>ACCEPT</b> number of infections decrease</p> <p><b>ACCEPT</b> herd immunity</p> | (3)  |